Supplementary Material for:
Diffusion-based Frameworks for Unsupervised Speech Enhancement

A Average metrics on the test datasets

We provide the metrics, per noise type and also per noise type and SNR.

A1 Average metrics per noise type on the test datasets

Table 1: Average SI-SDR per noise type in WSJ-QUT matched conditions.

Metrics			SI-SDR		
Noise types	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	Avg
Input	-1.8	-3.84	-2.63	-2.23	-2.6
SGMSE+	6.57	11.69	7.5	8.42	8.53
Conv-TasNet	**10.91**	**15.43**	**12.1**	**12.18**	**12.63**
RemixIT	3.01	5.75	3.82	4.03	4.14
RVAE	2.81	9.9	4.82	7.35	6.22
`UDiffSE`	4.12	6.02	3.27	3.93	4.35
`UDiffSE+`	3.23	4.86	2.07	2.7	3.24
`DEPSE-IL`	3.33	4.58	2.16	2.84	3.25
`DEPSE-TL`	2.99	5.37	2.46	3.23	3.53
`DiffUSEEN`	3.96	8.88	3.85	4.78	5.37
`ParaDiffUSE-IN`	2.83	-0.06	2.74	2.53	2.02
`ParaDiffUSE-EN`	7.04	11.32	7.73	7.94	8.49

Table 2: Average ESTOI per noise type in WSJ-QUT matched conditions.

Metrics			ESTOI		
Noise types	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	Avg
Input	0.46	0.56	0.5	0.48	0.5
SGMSE+	0.71	0.84	0.75	0.73	0.76
Conv-TasNet	**0.78**	**0.9**	**0.82**	**0.81**	**0.83**
RemixIT	0.63	0.76	0.67	0.66	0.68
RVAE	0.56	0.74	0.63	0.62	0.64
`UDiffSE`	0.59	0.69	0.6	0.59	0.61
`UDiffSE+`	0.56	0.64	0.56	0.55	0.58
`DEPSE-IL`	0.57	0.65	0.58	0.57	0.59
`DEPSE-TL`	0.58	0.67	0.6	0.59	0.61
`DiffUSEEN`	0.63	0.75	0.64	0.64	0.67
`ParaDiffUSE-IN`	0.55	0.54	0.58	0.55	0.56
`ParaDiffUSE-EN`	0.69	0.83	0.72	0.7	0.73

Table 3: Average DNS-MOS per noise type in WSJ-QUT matched conditions.

Metrics	DNS-MOS				
Noise types	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	Avg
Input	2.54	3.12	2.64	2.48	2.69
SGMSE+	**3.53**	**3.89**	**3.61**	**3.51**	**3.63**
Conv-TasNet	3.41	3.86	3.51	3.49	3.57
RemixIT	2.88	3.35	3.03	2.98	3.06
RVAE	3.12	3.54	3.17	3.18	3.25
`UDiffSE`	2.98	3.35	3.02	2.92	3.06
`UDiffSE+`	3.13	3.43	3.14	3.07	3.19
`DEPSE-IL`	3.02	3.42	3.07	2.96	3.11
`DEPSE-TL`	2.92	3.41	2.98	2.86	3.04
`DiffUSEEN`	3.07	3.46	3.11	3.01	3.16
`ParaDiffUSE-IN`	3.24	3.34	3.22	3.17	3.24
`ParaDiffUSE-EN`	3.28	3.84	3.37	3.29	3.44

Table 4: Average SI-SDR per noise type in WSJ-QUT mismatched conditions.

Metrics	SI-SDR				
Noise types	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	Avg
Input	-1.8	-3.84	-2.63	-2.23	-2.6
SGMSE+	0.92	5.66	2.74	2.91	3.02
Conv-TasNet	**4.08**	7.99	**5.05**	**6.2**	**5.83**
RemixIT	2.04	4.87	3.3	3.28	3.35
RVAE	2.13	**8.37**	3.51	5.88	4.98
`UDiffSE`	2.3	1.7	1.29	1.3	1.66
`UDiffSE+`	1.06	-0.18	-0.4	-0.33	0.06
`DEPSE-IL`	1.29	-0.37	-0.14	0.08	0.24
`DEPSE-TL`	2.52	3.99	1.84	2.24	2.66
`DiffUSEEN`	3.48	7.24	3.32	4.3	4.59
`ParaDiffUSE-IN`	0.43	-0.27	-0.15	-1.51	-0.39
`ParaDiffUSE-EN`	2.8	5.86	3.77	2.33	3.65

Table 5: Average ESTOI per noise type in WSJ-QUT mismatched conditions.

Metrics	STOI				
Noise types	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	Avg
Input	0.46	0.56	0.5	0.48	0.5
SGMSE+	0.5	0.73	0.62	0.61	0.61
Conv-TasNet	0.57	0.71	0.62	**0.63**	0.63
RemixIT	0.59	**0.74**	**0.65**	**0.63**	**0.65**
RVAE	0.53	0.68	0.59	0.57	0.59
`UDiffSE`	0.55	0.59	0.56	0.54	0.56
`UDiffSE+`	0.52	0.56	0.52	0.5	0.52
`DEPSE-IL`	0.53	0.57	0.53	0.51	0.53
`DEPSE-TL`	0.57	0.63	0.59	0.57	0.59
`DiffUSEEN`	**0.61**	0.71	0.62	0.62	0.64
`ParaDiffUSE-IN`	0.48	0.53	0.51	0.48	0.5
`ParaDiffUSE-EN`	0.53	0.69	0.61	0.58	0.6

Table 6: Average DNS-MOS per noise type in WSJ-QUT mismatched conditions.

Metrics	DNS-MOS				
Noise types	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	Avg
Input	2.54	3.12	2.64	2.48	2.69
SGMSE+	**3.48**	**3.75**	**3.61**	**3.55**	**3.59**
Conv-TasNet	3.0	3.21	3.04	3.03	3.07
RemixIT	2.81	3.21	2.93	2.84	2.94
RVAE	3.0	3.35	3.05	3.03	3.1
`UDiffSE`	3.03	3.29	3.03	2.94	3.07
`UDiffSE+`	3.1	3.35	3.07	3.01	3.13
`DEPSE-IL`	2.99	3.34	3.0	2.9	3.05
`DEPSE-TL`	2.95	3.39	2.99	2.89	3.05
`DiffUSEEN`	3.16	3.51	3.16	3.1	3.23
`ParaDiffUSE-IN`	3.09	3.34	3.14	3.12	3.17
`ParaDiffUSE-EN`	3.06	3.57	3.25	3.21	3.27

Table 7: Average SI-SDR per noise type in VB-DMD matched conditions.

Metrics	SI-SDR					
Noise types	bus	cafe	living	office	psquare	Avg
Input	8.35	8.5	8.47	8.46	8.45	8.45
SGMSE+	17.86	15.37	15.86	19.25	17.49	17.16
Conv-TasNet	**19.46**	**17.58**	**17.94**	**21.56**	**19.75**	**19.26**
RemixIT	2.11	1.76	1.25	0.16	2.62	1.58
RVAE	18.0	14.22	16.46	20.02	16.45	17.03
`UDiffSE`	11.72	13.02	13.04	15.86	14.47	13.62
`UDiffSE+`	9.09	10.29	10.25	11.46	10.9	10.4
`DEPSE-IL`	9.0	10.63	10.51	11.16	11.12	10.48
`DEPSE-TL`	11.11	13.4	13.67	16.33	14.68	13.84
`DiffUSEEN`	11.26	13.68	14.11	17.51	15.01	14.32
`ParaDiffUSE-IN`	13.24	13.2	13.33	12.96	14.99	13.55
`ParaDiffUSE-EN`	18.14	16.35	16.79	19.94	18.04	17.85

Table 8: Average ESTOI per noise type in VB-DMD matched conditions.

Metrics	STOI					
Noise types	bus	cafe	living	office	psquare	Avg
Input	0.86	0.69	0.72	0.89	0.77	0.79
SGMSE+	**0.89**	0.79	**0.82**	**0.9**	**0.84**	**0.85**
Conv-TasNet	0.88	**0.8**	**0.82**	**0.9**	**0.84**	**0.85**
RemixIT	0.66	0.61	0.62	0.67	0.64	0.64
RVAE	0.86	0.72	0.79	0.88	0.79	0.81
`UDiffSE`	0.87	0.75	0.78	0.88	0.8	0.82
`UDiffSE+`	0.81	0.72	0.74	0.83	0.77	0.77
`DEPSE-IL`	0.83	0.73	0.75	0.84	0.78	0.78
`DEPSE-TL`	0.87	0.75	0.78	0.89	0.81	0.82
`DiffUSEEN`	0.87	0.77	0.79	0.89	0.82	0.83
`ParaDiffUSE-IN`	0.85	0.75	0.76	0.86	0.8	0.8
`ParaDiffUSE-EN`	0.88	0.79	0.81	0.89	0.83	0.84

Table 9: Average DNS-MOS per noise type in VB-DMD matched conditions.

Metrics	DNS-MOS					
Noise types	bus	cafe	living	office	psquare	Avg
Input	3.27	2.87	2.82	3.32	3.08	3.07
SGMSE+	**3.5**	**3.49**	**3.52**	**3.51**	**3.51**	**3.51**
Conv-TasNet	3.35	3.24	3.27	3.38	3.32	3.31
RemixIT	2.93	2.89	2.85	2.9	2.92	2.9
RVAE	3.38	3.11	3.27	3.45	3.25	3.29
`UDiffSE`	3.42	3.2	3.19	3.45	3.3	3.31
`UDiffSE+`	3.32	3.2	3.21	3.35	3.25	3.27
`DEPSE-IL`	3.33	3.17	3.18	3.35	3.25	3.26
`DEPSE-TL`	3.4	3.21	3.2	3.45	3.31	3.31
`DiffUSEEN`	3.39	3.27	3.25	3.43	3.33	3.33
`ParaDiffUSE-IN`	3.39	3.23	3.26	3.44	3.33	3.33
`ParaDiffUSE-EN`	3.45	3.36	3.36	3.5	3.42	3.42

Table 10: Average SI-SDR per noise type in VB-DMD mismatched conditions.

Metrics	SI-SDR					
Noise types	bus	cafe	living	office	psquare	Avg
Input	8.35	8.5	8.47	8.46	8.45	8.45
SGMSE+	10.89	10.08	9.45	8.75	11.13	10.06
Conv-TasNet	10.8	11.93	11.38	9.03	12.51	11.13
RemixIT	1.38	1.48	1.15	-0.14	1.95	1.17
RVAE	13.53	10.28	12.45	**15.78**	11.85	12.77
`UDiffSE`	14.86	12.99	12.97	15.15	14.42	14.08
`UDiffSE+`	11.33	10.11	10.24	11.7	10.97	10.87
`DEPSE-IL`	11.5	10.57	10.57	11.75	11.32	11.14
`DEPSE-TL`	14.55	13.25	13.27	13.49	14.71	13.85
`DiffUSEEN`	**15.67**	**13.78**	**13.77**	14.46	**15.39**	**14.61**
`ParaDiffUSE-IN`	8.13	9.07	8.69	7.69	9.0	8.52
`ParaDiffUSE-EN`	10.63	11.21	10.64	8.49	12.55	10.71

Table 11: Average ESTOI per noise type in VB-DMD mismatched conditions.

Metrics	STOI					
Noise types	bus	cafe	living	office	psquare	Avg
Input	0.86	0.69	0.72	**0.89**	0.77	0.79
SGMSE+	**0.88**	0.75	0.77	**0.89**	0.81	0.82
Conv-TasNet	0.87	**0.78**	0.79	0.88	0.82	**0.83**
RemixIT	0.65	0.6	0.6	0.66	0.63	0.63
RVAE	0.8	0.67	0.75	0.84	0.73	0.76
`UDiffSE`	0.87	0.74	0.77	0.88	0.8	0.81
`UDiffSE+`	0.81	0.71	0.73	0.82	0.76	0.77
`DEPSE-IL`	0.82	0.72	0.74	0.84	0.77	0.78
`DEPSE-TL`	0.86	0.74	0.77	0.88	0.8	0.81
`DiffUSEEN`	0.86	0.75	0.77	0.88	0.81	0.82
`ParaDiffUSE-IN`	0.81	0.72	0.75	0.82	0.77	0.77
`ParaDiffUSE-EN`	0.87	0.77	**0.8**	**0.89**	**0.83**	**0.83**

Table 12: Average DNS-MOS per noise type in VB-DMD mismatched conditions.

Metrics	DNS-MOS					
Noise types	bus	cafe	living	office	psquare	Avg
Input	3.27	2.87	2.82	3.32	3.08	3.07
SGMSE+	**3.43**	3.21	3.12	3.42	3.32	3.3
Conv-TasNet	3.37	3.2	3.14	3.34	3.3	3.27
RemixIT	2.91	2.85	2.79	2.87	2.85	2.86
RVAE	3.36	3.15	**3.25**	3.4	3.26	3.28
`UDiffSE`	3.4	3.17	3.12	**3.46**	3.28	3.29
`UDiffSE+`	3.32	3.18	3.19	3.36	3.24	3.26
`DEPSE-IL`	3.3	3.16	3.13	3.34	3.21	3.23
`DEPSE-TL`	3.38	3.18	3.14	3.43	3.28	3.28
`DiffUSEEN`	3.36	3.22	3.17	3.43	3.3	3.3
`ParaDiffUSE-IN`	3.33	3.15	3.21	3.35	3.27	3.26
`ParaDiffUSE-EN`	**3.43**	**3.27**	**3.25**	3.45	**3.36**	**3.35**

B Investigating which mismatch is worse

In this section, we propose to check how mismatch in speech or/and in noise affects the `ParaDiffUSE-EN` performance on the WSJ0-QUT and VoiceBank-DEMAND(VB-DMD) test set. To do so, we consider the `ParaDiffUSE-EN` in its separated version. Given a test data set, we evaluate `ParaDiffUSE-EN` on that test set with four different setting:

- Matching with respect to speech and noise (full-matching): evaluate by using the checkpoint of diffusion model that is trained on the speech data set that matches the speech in the test set and the checkpoint of the noise diffusion model trained on noise data set that matches the noise in the test set.
- Mismatching with respect to speech and matching for the noise: evaluate by using the checkpoint of diffusion model that is trained on a speech data set that does not match the speech in the test set and the checkpoint of the noise diffusion model trained on noise data set that matches the noise in the test set.
- Matching with respect to speech and mismatching for the noise: evaluate by using the checkpoint of diffusion model that is trained on a speech data set that matches the speech in the test set and the checkpoint of the noise diffusion model trained on noise data set that does not match the noise in the test set.
- Mismatching with respect to speech and matching for the noise (full-mismatching): evaluate by using the checkpoint of diffusion model that is trained on a speech data set that does not match the speech in the test set and the checkpoint of the noise diffusion model trained on noise data set that does not match the noise in the test set.

The results are presented in tables 13 and 14. We can see that regarding the evaluation on VB-DMD (table 14), when using the noise diffusion model trained on QUT instead of the checkpoint trained on DMD, while using the speech diffusion trained on VB, we notice a huge degradation of the performance around 44% in terms of SI-SDR (17.32 → 9.66) with respect to the full matching scenario. On the evaluation of WSJ0-QUT test (table 13), the scenario of matching with respect to speech and mismatch for the noise results in a drop of 29.5% (9.19 → 6.47) with respect to the full matching situation. We conjecture this illustrates that the training noises in DEMAND are well diversified than that of the QUT training noise set so that, the checkpoint trained on DEMAND hurts less the performance on WSJ0-QUT when we have only noise mismatch, while as the training noise in QUT is less diversified, the performance on VB-DMD test set is more impacted when we have only noise mismatch.

We also note that the scenario of only speech speech mismatch and only noise mismatch seems to affect in the same range the SI-SDR on WSJ0-QUT test set. But under the hood, the mismatch in noise affects the ability to remove noise (i.e. there are more interference remaining after processing) which translates into lower SI-SIR compared to the scenario of only speech mismatch (13.72 vs 25.39). We observe the same thing on VB-DMD but in a more pronounced way: the SI-SIR is on average 11.83 dB in only noise mismatch scenario but is on average 34.49 dB in only speech mismatch. With respect to the full matching scenario in the WSJ0-QUT test set, the PESQ seems to be less affected by noise mismatch (2.47) than by speech mismatch (23.1) while the overall quality indicated by the DNS-MOS (P.808) is less affected by speech mismatch (3.44) than noise mismatch (3.25). We don't however see significantly such clean patterns on VB-DMD for PESQ and DNS-MOS.

The choice of the separated `ParaDiffUSE-EN` is motivated by...

Table 13: Average performance of separated `ParaDiffUSE-EN` on WSJ0-QUT test set in different configurations.

	SI-SDR	SI-SIR	SI-SAR	PESQ	STOI	DNS-MOS
Input	-2.60 ± 0.16	-2.60 ± 0.16	46.66 ± 0.43	1.83 ± 0.02	0.50 ± 0.01	2.69 ± 0.01
speech_WSJ0_noise_QUT_full_match	9.19 ± 0.16	24.63 ± 0.25	9.41 ± 0.16	2.68 ± 0.02	0.76 ± 0.01	3.50 ± 0.01
speech_WSJ0_noise_DMD_only_noise_mismatch	6.47 ± 0.25	13.72 ± 0.4	8.40 ± 0.19	2.47 ± 0.02	0.68 ± 0.01	3.25 ± 0.02
speech_VB_noise_QUT_only_speech_mismatch	6.72 ± 0.15	25.39 ± 0.22	6.82 ± 0.15	2.31 ± 0.02	0.66 ± 0.01	3.44 ± 0.01
speech_VB_noise_DMD_full_mismatch	3.62 ± 0.29	11.05 ± 0.44	5.42 ± 0.24	2.28 ± 0.02	0.60 ± 0.01	3.23 ± 0.02

Table 14: Average performance of separated `ParaDiffUSE-EN` on VB-DMD test set in different configurations.

	SI-SDR	SI-SIR	SI-SAR	PESQ	STOI	DNS-MOS
Input	8.45 ± 0.20	8.45 ± 0.20	47.51 ± 0.36	3.02 ± 0.02	0.79 ± 0.01	3.07 ± 0.01
speech_VB_noise_DMD_full_match	17.32 ± 0.11	28.48 ± 0.18	17.9 ± 0.11	3.50 ± 0.01	0.84 ± 0.00	3.43 ± 0.01
speech_VB_noise_QUT_only_noise_mismatch	9.66 ± 0.18	11.83 ± 0.22	16.0 ± 0.15	3.36 ± 0.01	0.82 ± 0.00	3.38 ± 0.01
speech_WSJ0_noise_DMD_only_speech_mismatch	16.51 ± 0.10	34.49 ± 0.29	16.67 ± 0.09	3.34 ± 0.02	0.82 ± 0.00	3.37 ± 0.01
speech_WSJ0_noise_QUT_full_mismatch	10.59 ± 0.19	12.98 ± 0.23	16.66 ± 0.17	3.35 ± 0.01	0.83 ± 0.00	3.35 ± 0.01

C Performance–speed trade-off of `DiffUSEEN` and `DEPSE-TL`

In this section, we propose to follow how the performances of `DiffUSEEN` and `DEPSE-TL` evolve in function of the number of reverse diffusion iterations in the mismatch scenarios for WSJ0-QUT and VB-DMD test sets. This experiment is conducted on an Nvidia Tesla T4 GPU (15 GiB VRAM). We can see that in the case of mismatch WSJ0-QUT, the SI-SDR, PESQ and ESTOI of `DiffUSEEN` are always greater than or equal to those of `DEPSE-TL` (except for 10 reverse diffusion iterations). For mismatch VB-DMD, the decrease in `DiffUSEEN`' SI-SDR and PESQ after 30 iterations leads to a slight under performance of `DiffUSEEN` with respect to `DEPSE-TL`. As the number of reverse iterations increases the real time factor also increases but this does not necessarily translate into an important improvement in terms of PESQ and ESTOI, for both methods.

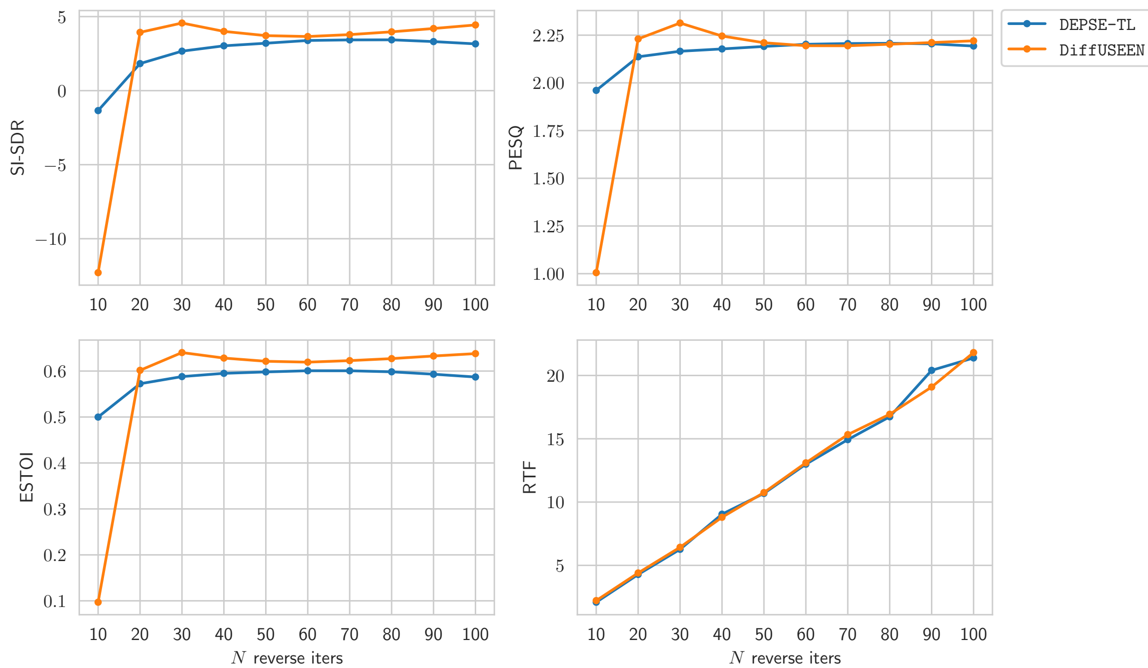

Figure 1: Performance on WSJ0-QUT test set in mismatched conditions and inference speed as a function of the number of reverse diffusion steps N

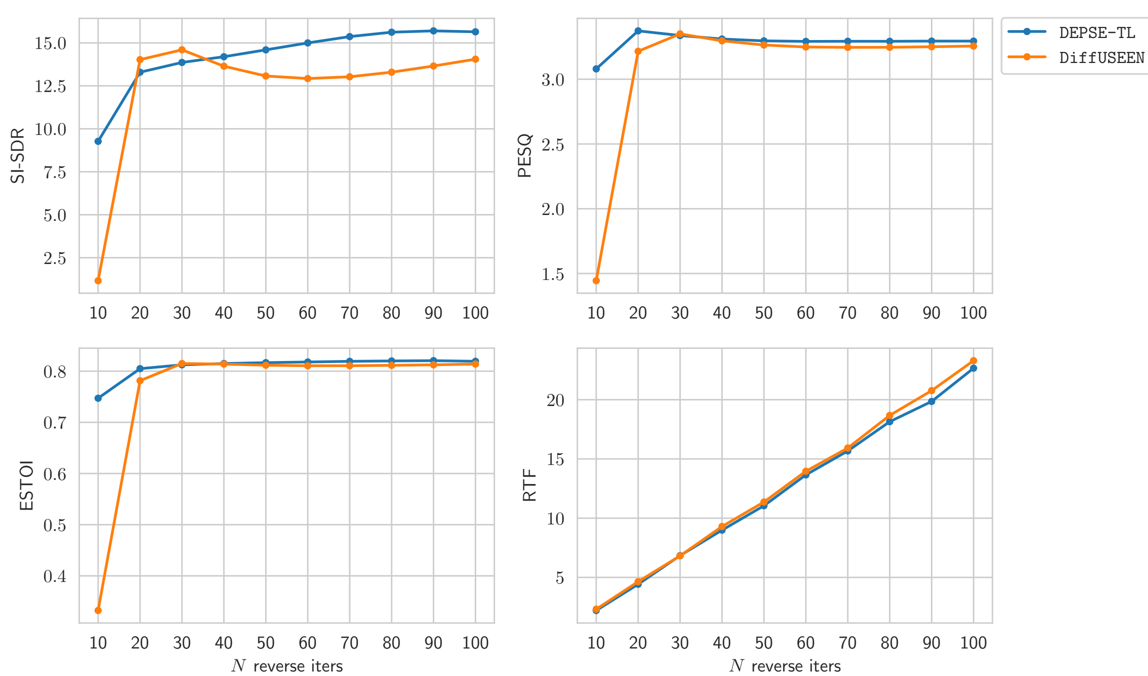

Figure 2: Performance on VB-DMD test set in mismatched conditions and inference speed as a function of the number of reverse diffusion steps N

D Distribution of the metrics in matched vs mismatched scenarios

We plot here violin plots of DNS-MOS, ESTOI, PESQ and SI-SDR distributions for different methods on the WSJ0-QUT and VB-DMD tests set in both matched and mismatched conditions. Similar matched and mismatched distributions for a given method indicate robustness to distribution shift in unseen test conditions and, thus, greater generalization capacity (at least on the datasets used here). A similar plot (Figure 3 in the main paper) is analyzed in subsection VI-B-2 of the main paper.

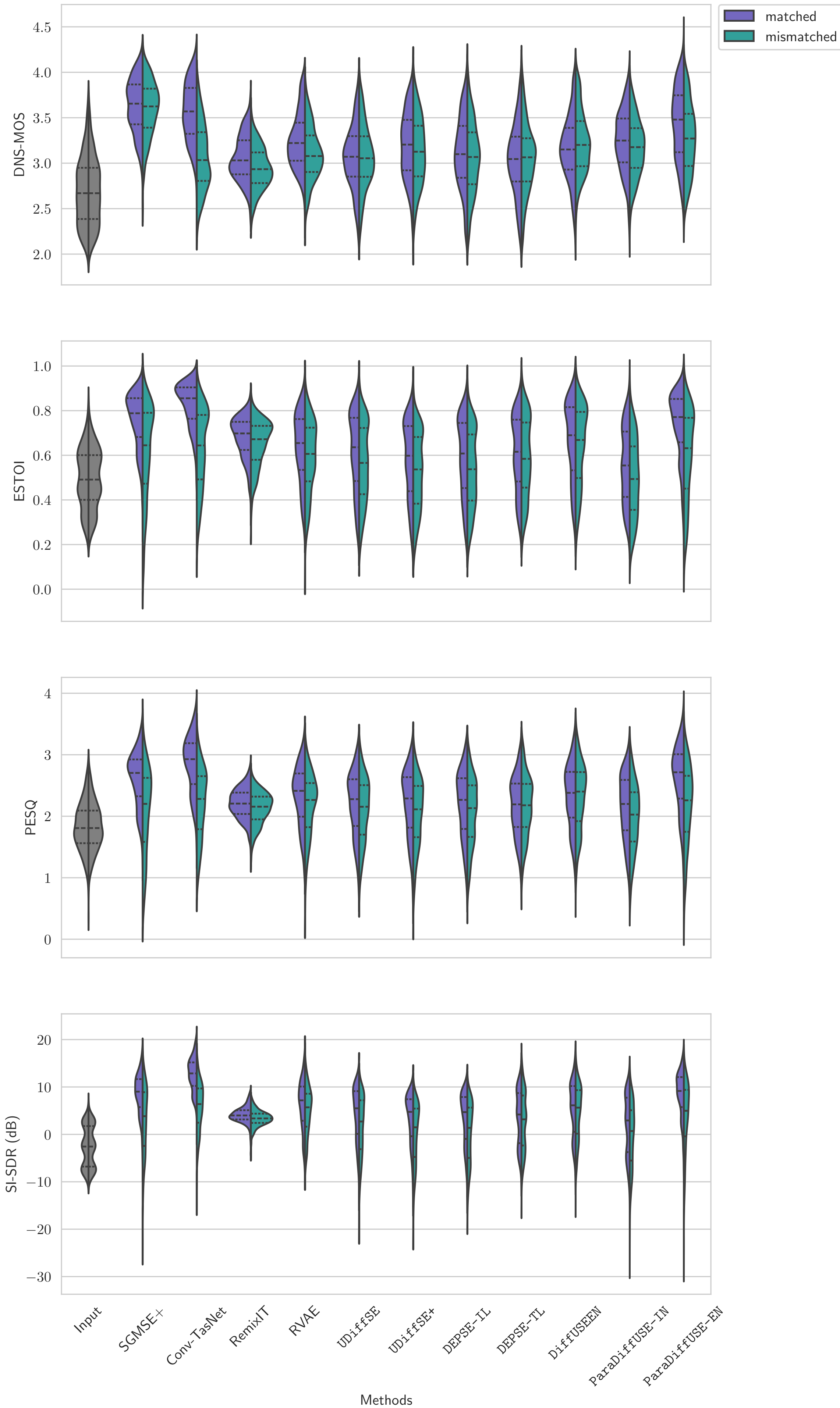

Figure 3: Violin plots showing the DNS-MOS, ESTOI, PESQ and SI-SDR distributions for the matched and mismatched conditions on the WSJ0-QUT test set, with dashed and dotted lines indicating the median and quartiles, respectively.

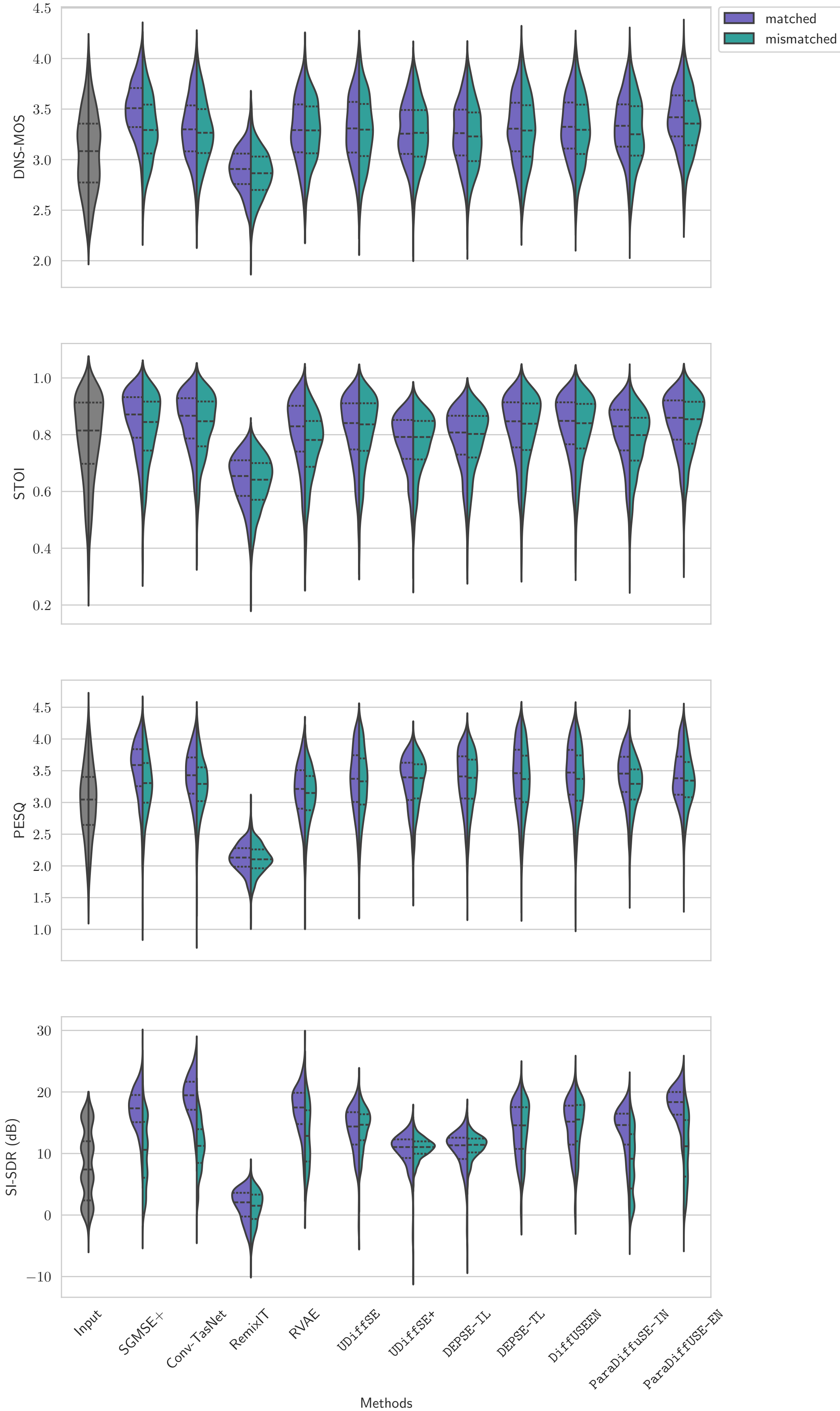

Figure 4: Violin plots showing the DNS-MOS, ESTOI, PESQ and SI-SDR distributions for the matched and mismatched conditions on the VB-DMD test set, with dashed and dotted lines indicating the median and quartiles, respectively.

E Average metrics per noise type and per signal-to-noise ratio (SNR) on the test datasets

Table 15: Average performance per noise type and SNR in WSJ-QUT matched conditions. Best results per column (per noise type) are in bold.

Noise type	Method	SI-SDR ↑				ESTOI ↑				DNS-MOS ↑			
		-5	**0**	**5**	**Avg**	**-5**	**0**	**5**	**Avg**	**-5**	**0**	**5**	**Avg**
CAFE-CAFE-2	Input	-6.68	-1.40	3.36	-1.80	0.31	0.46	0.62	0.46	2.33	2.52	2.79	2.54
	SGMSE+	1.82	7.32	11.09	6.57	0.57	0.74	0.85	0.71	**3.37**	**3.54**	**3.71**	**3.53**
	Conv-TasNet	**6.99**	**11.43**	**14.78**	**10.91**	**0.66**	**0.80**	**0.89**	**0.78**	3.15	3.44	3.67	3.41
	RemixIT	2.26	3.46	3.31	3.01	0.54	0.65	0.71	0.63	2.76	2.91	2.99	2.88
	RVAE	-2.71	3.63	8.16	2.81	0.39	0.58	0.73	0.56	3.00	3.11	3.26	3.12
	UDiffSE	-2.68	5.68	9.99	4.12	0.37	0.62	0.79	0.59	2.72	3.00	3.24	2.98
	UDiffSE+	-2.99	5.01	8.13	3.23	0.35	0.59	0.76	0.56	2.82	3.14	3.49	3.13
	DEPSE-IL	-3.24	5.08	8.67	3.33	0.35	0.60	0.77	0.57	2.71	3.02	3.37	3.02
	DEPSE-TL	-3.95	4.11	9.59	2.99	0.37	0.61	0.79	0.58	2.63	2.96	3.19	2.92
	DiffUSEEN	-2.79	5.14	10.24	3.96	0.42	0.68	0.83	0.63	2.82	3.08	3.33	3.07
	ParaDiffUSE-IN	-4.43	4.14	9.55	2.83	0.35	0.58	0.77	0.55	3.03	3.27	3.46	3.24
	ParaDiffUSE-EN	1.42	8.02	12.30	7.04	0.50	0.72	0.85	0.69	2.99	3.32	3.56	3.28
CAR-WINUPB-2	Input	-8.26	-3.41	1.52	-3.84	0.46	0.58	0.66	0.56	2.93	3.17	3.29	3.12
	SGMSE+	8.87	12.27	14.75	11.69	0.79	0.86	0.89	0.84	**3.81**	**3.94**	3.93	**3.89**
	Conv-TasNet	**13.05**	**15.89**	**18.05**	**15.43**	**0.86**	**0.92**	**0.93**	**0.90**	3.73	3.92	**3.96**	3.86
	RemixIT	7.22	5.50	4.09	5.75	0.75	0.78	0.75	0.76	3.33	3.38	3.33	3.35
	RVAE	7.02	10.24	13.32	9.90	0.64	0.77	0.83	0.74	3.34	3.60	3.75	3.54
	UDiffSE	1.33	7.03	11.04	6.02	0.55	0.73	0.82	0.69	3.12	3.39	3.61	3.35
	UDiffSE+	0.61	6.02	9.14	4.86	0.50	0.69	0.79	0.64	3.18	3.52	3.66	3.43
	DEPSE-IL	-0.41	5.98	9.56	4.58	0.50	0.70	0.80	0.65	3.15	3.52	3.67	3.42
	DEPSE-TL	-0.33	6.58	11.52	5.37	0.52	0.71	0.82	0.67	3.18	3.46	3.64	3.41
	DiffUSEEN	4.49	10.00	13.39	8.88	0.60	0.81	0.87	0.75	3.20	3.51	3.72	3.46
	ParaDiffUSE-IN	-6.75	1.05	7.51	-0.06	0.40	0.57	0.69	0.54	3.08	3.44	3.56	3.34
	ParaDiffUSE-EN	8.72	11.77	14.24	11.32	0.76	0.85	0.89	0.83	3.68	3.93	3.95	3.84
HOME-LIVINGB-2	Input	-7.18	-2.23	2.55	-2.63	0.38	0.49	0.64	0.50	2.49	2.63	2.82	2.64
	SGMSE+	3.22	8.24	11.98	7.50	0.65	0.76	0.86	0.75	**3.51**	**3.62**	**3.71**	**3.61**
	Conv-TasNet	**8.80**	**12.57**	**15.66**	**12.10**	**0.73**	**0.83**	**0.90**	**0.82**	3.35	3.53	3.70	3.51
	RemixIT	3.66	4.18	3.64	3.82	0.62	0.69	0.72	0.67	2.94	3.05	3.11	3.03
	RVAE	0.03	5.57	9.94	4.82	0.50	0.65	0.78	0.63	3.00	3.20	3.35	3.17
	UDiffSE	-2.83	4.49	9.52	3.27	0.44	0.62	0.78	0.60	2.81	3.05	3.25	3.02
	UDiffSE+	-3.94	3.71	7.77	2.07	0.40	0.59	0.75	0.56	2.87	3.20	3.40	3.14
	DEPSE-IL	-3.98	3.71	8.10	2.16	0.41	0.60	0.76	0.58	2.84	3.11	3.33	3.07
	DEPSE-TL	-3.84	3.51	9.14	2.46	0.44	0.62	0.78	0.60	2.77	3.00	3.22	2.98
	DiffUSEEN	-2.09	4.96	9.99	3.85	0.48	0.67	0.81	0.64	2.90	3.15	3.32	3.11
	ParaDiffUSE-IN	-3.28	3.62	9.23	2.74	0.43	0.58	0.76	0.58	2.98	3.24	3.50	3.22
	ParaDiffUSE-EN	3.22	8.23	12.77	7.73	0.58	0.75	0.86	0.72	3.16	3.40	3.59	3.37
STREET-KG-2	Input	-7.35	-2.45	2.76	-2.23	0.32	0.47	0.63	0.48	2.26	2.46	2.72	2.48
	SGMSE+	4.34	8.87	11.86	8.42	0.59	0.75	0.85	0.73	**3.32**	3.50	**3.71**	**3.51**
	Conv-TasNet	**8.57**	**12.44**	**15.35**	**12.18**	**0.71**	**0.82**	**0.90**	**0.81**	3.23	**3.52**	**3.71**	3.49
	RemixIT	4.32	4.35	3.51	4.03	0.59	0.67	0.72	0.66	2.86	3.02	3.06	2.98
	RVAE	2.30	8.05	11.51	7.35	0.43	0.64	0.78	0.62	2.91	3.16	3.45	3.18
	UDiffSE	-3.45	5.30	9.72	3.93	0.36	0.60	0.78	0.59	2.54	2.95	3.24	2.92
	UDiffSE+	-4.18	4.35	7.80	2.70	0.33	0.57	0.74	0.55	2.67	3.08	3.45	3.07
	DEPSE-IL	-4.27	4.35	8.26	2.84	0.35	0.58	0.76	0.57	2.52	2.99	3.34	2.96
	DEPSE-TL	-4.29	4.12	9.52	3.23	0.37	0.60	0.78	0.59	2.46	2.89	3.22	2.86
	DiffUSEEN	-2.51	6.05	10.58	4.78	0.42	0.67	0.82	0.64	2.64	3.06	3.33	3.01
	ParaDiffUSE-IN	-5.30	3.23	9.30	2.53	0.35	0.54	0.75	0.55	2.82	3.17	3.49	3.17
	ParaDiffUSE-EN	2.32	8.63	12.63	7.94	0.51	0.74	0.85	0.70	2.90	3.33	3.62	3.29

Table 16: Average performance per noise type and SNR in WSJ-QUT mismatched conditions. Best results per column (per noise type) are in bold.

Noise type	**Method**	**SI-SDR ↑**				**ESTOI ↑**				**DNS-MOS ↑**			
		-5	**0**	**5**	**Avg**	**-5**	**0**	**5**	**Avg**	**-5**	**0**	**5**	**Avg**
CAFE-CAFE-2	Input	-6.68	-1.40	3.36	-1.80	0.31	0.46	0.62	0.46	2.33	2.52	2.79	2.54
	SGMSE+	-7.81	2.63	8.84	0.92	0.25	0.53	0.76	0.50	**3.22**	**3.52**	**3.74**	**3.48**
	Conv-TasNet	-2.04	**5.02**	**9.95**	**4.08**	0.35	0.59	0.79	0.57	2.67	3.01	3.36	3.00
	RemixIT	**0.74**	2.48	2.98	2.04	**0.49**	0.61	0.69	0.59	2.69	2.84	2.93	2.81
	RVAE	-3.38	2.96	7.45	2.13	0.36	0.54	0.71	0.53	2.87	3.00	3.17	3.00
	`UDiffSE`	-4.81	3.93	8.43	2.30	0.34	0.58	0.76	0.55	2.76	3.03	3.34	3.03
	`UDiffSE+`	-5.95	2.99	6.68	1.06	0.31	0.54	0.72	0.52	2.79	3.10	3.48	3.10
	`DEPSE-IL`	-5.68	2.98	7.17	1.29	0.32	0.55	0.73	0.53	2.67	2.98	3.36	2.99
	`DEPSE-TL`	-4.28	3.63	8.95	2.52	0.36	0.60	0.77	0.57	2.68	2.99	3.21	2.95
	`DiffUSEEN`	-3.08	4.67	9.55	3.48	0.40	**0.65**	**0.81**	**0.61**	2.92	3.17	3.42	3.16
	`ParaDiffUSE-IN`	-6.75	1.72	7.06	0.43	0.28	0.50	0.69	0.48	2.94	3.12	3.22	3.09
	`ParaDiffUSE-EN`	-4.94	4.48	9.61	2.80	0.30	0.57	0.75	0.53	2.81	3.10	3.29	3.06
CAR-WINUPB-2	Input	**5.74**	-3.41	1.52	-3.84	0.46	0.58	0.66	0.56	2.93	3.17	3.29	3.12
	SGMSE+	-1.33	8.02	12.15	5.66	0.58	**0.79**	0.84	0.73	**3.57**	**3.83**	**3.90**	**3.75**
	Conv-TasNet	3.60	**9.14**	**12.48**	7.99	0.56	0.76	0.84	0.71	2.81	3.30	3.65	3.21
	RemixIT	**5.74**	4.82	3.76	4.87	**0.71**	0.76	0.74	**0.74**	3.17	3.26	3.20	3.21
	RVAE	5.32	8.98	11.70	**8.37**	0.57	0.71	0.79	0.68	3.13	3.39	3.58	3.35
	`UDiffSE`	-3.16	1.69	8.14	1.70	0.46	0.59	0.76	0.59	3.05	3.29	3.62	3.29
	`UDiffSE+`	-5.45	0.15	6.42	-0.18	0.43	0.57	0.73	0.56	3.12	3.38	3.63	3.35
	`DEPSE-IL`	-6.11	0.02	6.78	-0.37	0.43	0.57	0.74	0.57	3.11	3.37	3.62	3.34
	`DEPSE-TL`	-1.80	3.85	11.84	3.99	0.48	0.64	0.83	0.63	3.15	3.40	3.69	3.39
	`DiffUSEEN`	2.84	8.01	12.17	7.24	0.56	0.76	**0.86**	0.71	3.21	3.61	3.79	3.51
	`ParaDiffUSE-IN`	-4.59	0.38	4.69	-0.27	0.42	0.54	0.65	0.53	3.15	3.39	3.51	3.34
	`ParaDiffUSE-EN`	1.30	6.02	11.72	5.86	0.56	0.72	0.84	0.69	3.31	3.63	3.82	3.57
HOME-LIVINGB-2	Input	-7.18	-2.23	2.55	-2.63	0.38	0.49	0.64	0.50	2.49	2.63	2.82	2.64
	SGMSE+	-4.32	4.36	9.75	2.74	0.42	0.66	**0.82**	0.62	**3.41**	**3.67**	**3.78**	**3.61**
	Conv-TasNet	-0.65	**6.51**	10.54	**5.05**	0.44	0.65	0.81	0.62	2.76	3.06	3.37	3.04
	RemixIT	**2.94**	3.66	3.36	3.30	**0.58**	**0.67**	0.71	**0.65**	2.84	2.93	3.03	2.93
	RVAE	-1.19	4.08	8.69	3.51	0.45	0.60	0.74	0.59	2.87	3.06	3.27	3.05
	`UDiffSE`	-5.10	2.25	8.16	1.29	0.41	0.57	0.75	0.56	2.82	3.03	3.29	3.03
	`UDiffSE+`	-7.05	1.00	6.33	-0.40	0.37	0.52	0.71	0.52	2.81	3.10	3.36	3.07
	`DEPSE-IL`	-6.64	0.98	6.68	-0.14	0.38	0.53	0.72	0.53	2.77	3.01	3.28	3.00
	`DEPSE-TL`	-4.55	2.50	9.02	1.84	0.43	0.59	0.77	0.59	2.77	3.03	3.24	2.99
	`DiffUSEEN`	-2.63	4.17	9.73	3.32	0.46	0.64	0.80	0.62	2.96	3.18	3.40	3.16
	`ParaDiffUSE-IN`	-7.00	0.70	7.38	-0.15	0.35	0.51	0.71	0.51	2.94	3.16	3.36	3.14
	`ParaDiffUSE-EN`	-2.83	4.76	**10.86**	3.77	0.43	0.63	0.80	0.61	2.99	3.32	3.50	3.25
STREET-KG-2	Input	-7.35	-2.45	2.76	-2.23	0.32	0.47	0.63	0.48	2.26	2.46	2.72	2.48
	SGMSE+	-5.20	4.20	9.45	2.91	0.38	0.65	0.80	0.61	**3.33**	**3.57**	**3.74**	**3.55**
	Conv-TasNet	0.98	**6.94**	10.50	**6.20**	0.43	0.64	0.80	0.63	2.71	2.98	3.38	3.03
	RemixIT	2.97	3.75	3.19	3.28	**0.53**	**0.64**	0.70	0.63	2.70	2.87	2.97	2.84
	RVAE	0.91	6.57	9.97	5.88	0.36	0.58	0.74	0.57	2.74	3.01	3.30	3.03
	`UDiffSE`	-6.77	2.22	8.11	1.30	0.33	0.53	0.74	0.54	2.56	2.95	3.29	2.94
	`UDiffSE+`	-8.24	0.89	6.06	-0.33	0.30	0.49	0.70	0.50	2.63	2.99	3.39	3.01
	`DEPSE-IL`	-7.60	1.01	6.49	0.08	0.31	0.50	0.71	0.51	2.48	2.89	3.29	2.90
	`DEPSE-TL`	-5.72	2.93	9.13	2.24	0.36	0.57	0.77	0.57	2.49	2.90	3.25	2.89
	`DiffUSEEN`	-3.12	5.66	10.12	4.30	0.40	0.65	0.80	0.62	2.79	3.11	3.38	3.10
	`ParaDiffUSE-IN`	-10.14	-0.85	6.02	-1.51	0.29	0.46	0.67	0.48	2.85	3.10	3.39	3.12
	`ParaDiffUSE-EN`	-7.50	4.23	**10.50**	2.33	0.33	0.60	**0.80**	0.58	2.84	3.21	3.55	3.21

Table 17: Average performance per noise type and SNR in VB-DMD matched conditions. Best results per column (per noise type) are in bold.

Noise type	Method	SI-SDR ↑					ESTOI ↑					DNS-MOS ↑				
		2.5	7.5	12.5	17.5	Avg	2.5	7.5	12.5	17.5	Avg	2.5	7.5	12.5	17.5	Avg
BUS	Input	0.89	5.99	10.92	15.95	8.35	0.78	0.87	0.89	0.92	0.86	3.10	3.24	3.33	3.39	3.27
	SGMSE+	16.59	17.65	17.98	19.30	17.86	0.85	0.90	0.91	**0.93**	0.89	**3.50**	3.51	3.51	3.49	3.50
	Conv-TasNet	**17.93**	19.03	19.62	**21.36**	19.46	0.83	0.89	0.90	0.92	0.88	3.22	3.35	3.39	3.43	3.35
	RemixIT	0.65	1.28	2.72	3.85	2.11	0.63	0.67	0.66	0.67	0.66	2.86	2.91	2.95	2.99	2.93
	RVAE	16.09	18.48	17.76	19.77	18.00	0.80	0.87	0.87	0.89	0.86	3.31	3.38	3.39	3.44	3.38
	UDiffSE	4.55	11.46	14.33	16.86	11.72	0.80	0.87	0.89	0.91	0.87	3.32	3.42	3.46	3.47	3.42
	UDiffSE+	2.94	9.52	11.50	12.66	9.09	0.75	0.82	0.83	0.84	0.81	3.22	3.32	3.36	3.38	3.32
	DEPSE-IL	2.68	9.15	11.54	12.87	9.00	0.76	0.83	0.85	0.86	0.83	3.21	3.34	3.39	3.40	3.33
	DEPSE-TL	4.24	9.94	13.56	17.02	11.11	0.81	0.88	0.89	0.91	0.87	3.31	3.42	3.42	3.47	3.40
	DiffUSEEN	4.72	10.18	13.55	16.88	11.26	0.81	0.88	0.89	0.91	0.87	3.29	3.40	3.40	3.46	3.39
	ParaDiffUSE-IN	8.28	13.83	14.77	16.28	13.24	0.81	0.86	0.86	0.87	0.85	3.32	3.39	3.41	3.45	3.39
	ParaDiffUSE-EN	14.96	**18.85**	**19.16**	19.72	**18.14**	0.83	0.88	**0.89**	0.91	0.88	3.38	**3.47**	3.46	**3.50**	**3.45**
CAFE	Input	-0.43	1.25	2.59	3.65	1.76	0.51	0.66	0.75	0.83	0.69	2.49	2.77	2.99	3.24	2.87
	SGMSE+	10.54	14.64	16.76	19.63	15.37	0.67	0.79	0.82	0.88	0.79	3.38	3.52	3.51	3.55	3.49
	Conv-TasNet	12.93	16.61	19.12	21.74	17.58	**0.70**	0.79	0.83	0.88	0.80	2.95	3.23	3.33	3.46	3.24
	RemixIT	-0.43	1.25	2.59	3.65	1.76	0.55	0.60	0.63	0.67	0.61	2.74	2.84	2.93	3.04	2.89
	RVAE	8.90	13.30	16.10	18.68	14.22	0.58	0.71	0.78	0.84	0.72	2.90	3.06	3.17	3.33	3.11
	UDiffSE	8.86	12.13	14.54	16.63	13.02	0.61	0.73	0.79	0.85	0.75	2.91	3.13	3.28	3.47	3.20
	UDiffSE+	7.58	10.05	11.32	12.26	10.29	0.59	0.70	0.76	0.81	0.72	2.95	3.17	3.27	3.41	3.20
	DEPSE-IL	7.61	10.48	11.81	12.69	10.63	0.60	0.72	0.77	0.82	0.73	2.92	3.15	3.24	3.39	3.17
	DEPSE-TL	7.97	12.08	15.36	18.31	13.40	0.62	0.74	0.80	0.86	0.75	2.90	3.16	3.30	3.48	3.21
	DiffUSEEN	8.61	12.35	15.42	18.40	13.68	0.65	0.75	0.81	0.86	0.77	3.02	3.23	3.33	3.48	3.27
	ParaDiffUSE-IN	8.41	12.60	15.18	16.71	13.20	0.61	0.74	0.79	0.84	0.75	2.99	3.20	3.31	3.42	3.23
	ParaDiffUSE-EN	**12.06**	**15.72**	**17.91**	**19.80**	**16.35**	0.68	**0.78**	**0.81**	**0.86**	**0.79**	**3.18**	**3.37**	**3.41**	**3.50**	**3.36**
LIVING	Input	1.07	6.12	10.97	16.07	8.47	0.57	0.72	0.76	0.85	0.72	2.51	2.73	2.93	3.15	2.82
	SGMSE+	11.87	15.70	16.83	19.21	15.86	0.73	**0.83**	**0.84**	**0.89**	**0.82**	**3.46**	**3.54**	3.53	**3.55**	**3.52**
	Conv-TasNet	**14.00**	**17.56**	**18.96**	**21.40**	**17.94**	**0.74**	0.82	0.83	0.88	0.82	3.05	3.24	3.35	3.43	3.27
	RemixIT	-1.89	0.62	2.49	3.90	1.25	0.57	0.63	0.63	0.65	0.62	2.72	2.79	2.92	2.97	2.85
	RVAE	12.17	16.05	17.74	20.04	16.46	0.70	0.79	0.81	0.87	0.79	3.07	3.25	**3.35**	3.41	3.27
	UDiffSE	8.81	12.61	14.46	16.43	13.04	0.67	0.78	0.80	0.86	0.78	2.89	3.14	3.33	3.39	3.19
	UDiffSE+	7.55	10.14	11.25	12.16	10.25	0.64	0.74	0.76	0.81	0.74	3.00	3.16	3.32	3.38	3.21
	DEPSE-IL	7.31	10.60	11.68	12.52	10.51	0.64	0.76	0.77	0.83	0.75	2.92	3.16	3.28	3.37	3.18
	DEPSE-TL	8.64	12.66	15.40	18.20	13.67	0.68	0.79	0.81	0.87	0.78	2.92	3.17	3.33	3.39	3.20
	DiffUSEEN	9.79	13.19	15.46	18.21	14.11	0.70	0.80	0.81	0.87	0.79	3.04	3.23	3.35	3.39	3.25
	ParaDiffUSE-IN	8.65	13.73	14.97	16.09	13.33	0.65	0.77	0.79	0.85	0.76	3.05	3.24	3.35	3.41	3.26
	ParaDiffUSE-EN	12.79	16.58	18.17	19.76	16.79	0.73	0.82	0.83	0.87	0.81	3.21	3.35	**3.43**	3.45	3.36
OFFICE	Input	0.87	5.90	10.93	15.91	8.46	0.83	0.86	**0.91**	0.94	**0.89**	3.11	3.30	3.40	3.47	3.32
	SGMSE+	17.37	18.70	19.70	21.15	19.25	**0.87**	0.89	**0.91**	**0.94**	0.90	3.52	3.51	3.53	3.49	3.51
	Conv-TasNet	**19.69**	**20.86**	**22.06**	**23.57**	**21.56**	0.86	0.88	**0.91**	0.93	0.90	3.20	3.37	3.47	3.49	3.38
	RemixIT	-3.59	-0.86	1.78	3.16	0.16	0.65	0.66	0.67	0.69	0.67	2.70	2.87	2.98	3.05	2.90
	RVAE	18.47	19.44	20.81	21.31	20.02	0.84	0.86	0.89	0.91	0.88	3.36	3.46	3.48	3.49	3.45
	UDiffSE	10.90	15.39	18.03	18.97	15.86	0.85	0.86	0.90	0.93	0.88	3.32	3.45	3.50	3.52	3.45
	UDiffSE+	8.31	11.73	12.76	12.93	11.46	0.79	0.81	0.84	0.86	0.83	3.28	3.37	3.36	3.37	3.35
	DEPSE-IL	6.77	11.33	13.09	13.28	11.16	0.81	0.83	0.86	0.88	0.84	3.28	3.36	3.38	3.40	3.35
	DEPSE-TL	11.61	15.32	18.27	19.97	16.33	0.85	0.87	0.90	0.92	0.89	3.35	3.46	3.48	3.49	3.45
	DiffUSEEN	14.45	16.38	18.90	20.20	17.51	0.86	0.87	0.90	0.92	0.89	3.33	3.45	3.46	3.48	3.43
	ParaDiffUSE-IN	6.25	11.96	16.39	16.98	12.96	0.83	0.84	0.87	0.89	0.86	3.37	3.45	3.47	3.45	3.44
	ParaDiffUSE-EN	18.77	19.59	20.25	21.11	19.94	0.86	**0.88**	**0.91**	**0.93**	0.89	3.39	**3.49**	**3.55**	**3.55**	**3.50**
PSQUARE	Input	0.90	6.04	10.86	15.90	8.45	0.66	0.75	0.82	0.85	0.77	2.78	3.03	3.21	3.28	3.08
	SGMSE+	14.91	16.57	18.58	19.88	17.49	**0.78**	**0.83**	**0.87**	**0.88**	**0.84**	**3.42**	**3.55**	**3.55**	**3.51**	**3.51**
	Conv-TasNet	**16.94**	**18.65**	**20.95**	**22.42**	**19.75**	0.77	**0.83**	**0.87**	**0.88**	**0.84**	3.09	3.34	3.45	3.40	3.32
	RemixIT	0.24	2.47	3.45	4.30	2.62	0.59	0.64	0.65	0.67	0.64	2.83	2.92	2.98	2.95	2.92
	RVAE	13.91	15.46	17.58	18.81	16.45	0.72	0.78	0.83	0.84	0.79	3.12	3.24	3.34	3.31	3.25
	UDiffSE	10.71	13.88	16.02	17.23	14.47	0.73	0.79	0.84	0.86	0.80	3.06	3.28	3.44	3.41	3.30
	UDiffSE+	8.53	10.81	11.90	12.32	10.90	0.70	0.76	0.80	0.81	0.77	3.06	3.25	3.38	3.32	3.25
	DEPSE-IL	8.13	11.28	12.38	12.66	11.12	0.71	0.77	0.82	0.82	0.78	3.05	3.25	3.37	3.31	3.25
	DEPSE-TL	9.94	13.50	16.46	18.76	14.68	0.74	0.80	0.85	0.86	0.81	3.09	3.31	3.45	3.39	3.31
	DiffUSEEN	10.74	13.88	16.62	18.75	15.01	0.76	0.81	0.85	0.86	0.82	3.14	3.33	3.44	3.39	3.33
	ParaDiffUSE-IN	11.95	15.08	16.35	16.53	14.99	0.74	0.80	0.83	0.84	0.80	3.18	3.34	3.42	3.38	3.33
	ParaDiffUSE-EN	15.56	17.37	19.27	19.93	18.04	0.77	0.83	0.86	0.87	0.83	3.28	3.45	3.49	3.45	3.42

Table 18: Average performance per noise type and SNR in VB-DMD mismatched conditions. Best results per column (per noise type) are in bold.

Noise type	**Method**	**SI-SDR ↑**					**ESTOI ↑**					**DNS-MOS ↑**				
		2.5	**7.5**	**12.5**	**17.5**	**Avg**	**2.5**	**7.5**	**12.5**	**17.5**	**Avg**	**2.5**	**7.5**	**12.5**	**17.5**	**Avg**
BUS	Input	0.89	5.99	10.92	15.95	8.35	0.78	0.87	0.89	**0.92**	0.86	3.10	3.24	3.33	3.39	3.27
	SGMSE+	5.81	8.20	12.38	17.47	10.89	**0.81**	**0.88**	**0.90**	**0.92**	**0.88**	**3.35**	**3.43**	**3.45**	3.50	**3.43**
	Conv-TasNet	7.21	8.83	11.87	15.51	10.80	0.81	0.87	0.89	**0.92**	0.87	3.28	3.37	3.40	3.45	3.37
	RemixIT	-0.54	0.31	2.19	3.65	1.38	0.61	0.66	0.66	0.67	0.65	2.83	2.90	2.93	2.98	2.91
	RVAE	**12.02**	13.93	13.59	14.63	13.53	0.75	0.81	0.81	0.84	0.80	3.28	3.38	3.37	3.42	3.36
	UDiffSE	11.78	14.35	15.85	17.61	14.86	**0.81**	0.87	0.88	0.91	0.87	3.30	3.40	3.43	3.45	3.40
	UDiffSE+	10.68	11.40	11.32	11.94	11.33	0.77	0.82	0.82	0.83	0.81	3.22	3.31	3.36	3.39	3.32
	DEPSE-IL	10.22	11.73	11.66	12.44	11.50	0.77	0.83	0.84	0.86	0.82	3.19	3.30	3.34	3.37	3.30
	DEPSE-TL	9.74	13.52	16.29	18.88	14.55	0.80	0.87	0.88	0.90	0.86	3.27	3.38	3.39	3.46	3.38
	DiffUSEEN	11.89	**14.96**	**16.80**	**18.88**	**15.67**	**0.81**	0.87	0.88	0.90	0.86	3.27	3.37	3.38	3.44	3.36
	ParaDiffUSE-IN	1.34	5.90	10.59	15.01	8.13	0.77	0.82	0.82	0.84	0.81	3.25	3.35	3.35	3.39	3.33
	ParaDiffUSE-EN	4.87	8.20	12.49	17.28	10.63	**0.82**	**0.88**	**0.89**	0.91	**0.87**	3.34	**3.43**	**3.45**	**3.51**	**3.43**
CAFE	Input	1.05	6.11	10.91	16.05	8.50	0.51	0.66	0.75	0.83	0.69	2.49	2.77	2.99	3.24	2.87
	SGMSE+	3.64	7.94	12.17	16.69	10.08	0.61	0.74	0.79	0.86	0.75	2.94	3.18	3.29	3.44	3.21
	Conv-TasNet	7.89	10.92	12.90	16.07	11.93	0.66	0.77	0.81	0.87	0.78	2.94	3.19	3.27	3.42	3.20
	RemixIT	-0.91	0.93	2.41	3.55	1.48	0.53	0.58	0.63	0.66	0.60	2.67	2.80	2.93	3.02	2.85
	RVAE	6.27	9.61	12.06	13.28	10.28	0.51	0.65	0.72	0.79	0.67	2.96	3.13	3.18	3.34	3.15
	UDiffSE	**8.99**	12.13	14.48	16.44	12.99	0.61	0.72	0.79	0.85	0.74	2.87	3.12	3.26	3.46	3.17
	UDiffSE+	8.11	9.78	11.00	11.60	10.11	0.59	0.70	0.76	0.81	0.71	2.94	3.15	3.26	3.39	3.18
	DEPSE-IL	8.35	10.35	11.47	12.17	10.57	0.60	0.71	0.77	0.82	0.72	2.91	3.13	3.22	3.38	3.16
	DEPSE-TL	7.45	12.02	15.40	18.25	13.25	0.60	0.73	0.79	0.86	0.74	2.85	3.14	3.29	3.46	3.18
	DiffUSEEN	8.24	**12.78**	**15.70**	**18.50**	**13.78**	0.63	0.74	0.80	0.86	0.75	2.93	3.19	3.29	**3.47**	**3.22**
	ParaDiffUSE-IN	2.26	7.42	11.50	15.24	9.07	0.58	0.72	0.77	0.82	0.72	2.84	3.16	3.25	3.37	3.15
	ParaDiffUSE-EN	4.84	9.55	13.47	17.10	11.21	**0.65**	**0.77**	**0.81**	**0.87**	**0.77**	**3.02**	**3.27**	**3.32**	**3.48**	**3.27**
LIVING	Input	1.07	6.12	10.97	16.07	8.47	0.57	0.72	0.76	0.85	0.72	2.51	2.73	2.93	3.15	2.82
	SGMSE+	2.35	7.29	11.94	16.55	9.45	0.64	0.77	0.80	0.86	0.77	2.86	3.04	3.22	3.36	3.12
	Conv-TasNet	7.05	10.17	12.80	15.69	11.38	0.68	0.80	0.82	0.87	0.79	2.87	3.09	3.27	3.34	3.14
	RemixIT	-1.95	0.51	2.35	3.81	1.15	0.54	0.62	0.62	0.64	0.60	2.67	2.74	2.85	2.92	2.79
	RVAE	9.24	13.92	12.81	13.89	12.45	0.64	0.76	0.76	0.82	0.75	3.07	3.22	3.33	3.39	3.25
	UDiffSE	**8.87**	12.52	14.38	16.25	12.97	0.65	0.77	0.79	0.86	0.77	2.80	3.06	3.29	3.35	3.12
	UDiffSE+	8.08	10.34	10.86	11.74	10.24	0.63	0.74	0.75	0.81	0.73	2.98	3.14	3.29	3.34	3.19
	DEPSE-IL	8.12	10.69	11.34	12.18	10.57	0.63	0.75	0.76	0.82	0.74	2.89	3.06	3.26	3.32	3.13
	DEPSE-TL	7.41	12.25	15.43	18.21	13.27	0.65	0.77	0.80	0.86	0.77	2.85	3.09	3.28	3.34	3.14
	DiffUSEEN	8.37	**13.00**	**15.61**	**18.32**	**13.77**	0.66	0.78	0.80	0.86	0.77	2.92	3.15	3.29	3.33	3.17
	ParaDiffUSE-IN	1.42	6.92	11.43	15.31	8.69	0.63	0.76	0.77	0.83	0.75	2.95	3.20	3.33	3.39	3.21
	ParaDiffUSE-EN	3.62	8.77	13.31	17.19	10.64	**0.71**	**0.81**	**0.82**	**0.87**	**0.80**	**2.95**	**3.24**	**3.37**	**3.42**	**3.25**
OFFICE	Input	0.87	5.90	10.93	15.91	8.46	0.83	0.86	**0.91**	**0.94**	**0.89**	3.11	3.30	3.40	3.47	3.32
	SGMSE+	1.66	6.19	11.06	15.85	8.75	0.84	**0.87**	0.90	0.93	**0.89**	3.26	3.41	3.49	3.52	3.42
	Conv-TasNet	2.97	6.83	11.11	14.99	9.03	0.83	0.86	**0.91**	**0.94**	0.88	3.13	3.31	3.43	3.48	3.34
	RemixIT	-3.91	-1.27	1.48	3.02	-0.14	0.63	0.65	0.66	0.69	0.66	2.63	2.84	2.96	3.04	2.87
	RVAE	**15.12**	**16.24**	15.37	16.37	**15.78**	0.81	0.83	0.84	0.87	0.84	3.32	3.37	3.44	3.48	3.40
	UDiffSE	11.56	14.63	16.39	17.92	15.15	**0.85**	0.86	0.90	0.93	0.88	3.32	**3.47**	3.50	**3.53**	3.46
	UDiffSE+	10.65	11.68	12.12	12.31	11.70	0.80	0.81	0.83	0.85	0.82	3.31	3.36	3.38	3.38	3.36
	DEPSE-IL	10.13	11.92	12.30	12.60	11.75	0.81	0.82	0.85	0.87	0.84	3.26	3.34	3.36	3.38	3.34
	DEPSE-TL	7.30	12.09	15.80	18.55	13.49	**0.85**	0.86	0.90	0.92	0.88	3.32	3.44	3.48	**3.49**	3.43
	DiffUSEEN	8.79	13.42	**16.56**	**18.88**	14.46	**0.85**	**0.86**	0.89	0.92	0.88	**3.33**	3.44	3.45	3.48	3.43
	ParaDiffUSE-IN	0.60	5.52	10.17	14.22	7.69	0.80	0.80	0.83	0.86	0.82	3.27	3.37	3.38	3.38	3.35
	ParaDiffUSE-EN	1.32	6.07	10.89	15.42	8.49	0.85	**0.87**	**0.90**	**0.93**	**0.89**	3.30	3.45	**3.51**	**3.53**	3.45
PSQUARE	Input	0.90	6.04	10.86	15.90	8.45	0.66	0.75	0.82	0.85	0.77	2.78	3.03	3.21	3.28	3.08
	SGMSE+	5.32	9.42	12.90	16.78	11.13	0.73	0.80	0.84	0.86	0.81	3.15	3.31	3.41	3.41	3.32
	Conv-TasNet	9.34	11.63	13.37	15.65	12.51	0.75	**0.82**	**0.85**	**0.86**	0.82	3.12	3.31	3.40	3.37	3.30
	RemixIT	-0.79	1.62	2.90	4.05	1.95	0.58	0.63	0.64	0.66	0.63	2.72	2.84	2.93	2.93	2.85
	RVAE	9.58	11.74	13.03	13.02	11.85	0.64	0.73	0.78	0.78	0.73	3.14	3.26	3.33	3.29	3.26
	UDiffSE	**11.38**	13.74	15.71	16.79	14.42	0.72	0.79	0.84	0.85	0.80	3.04	3.26	3.41	3.40	3.28
	UDiffSE+	9.96	10.71	11.61	11.59	10.97	0.70	0.76	0.80	0.80	0.76	3.08	3.25	3.36	3.28	3.24
	DEPSE-IL	9.99	11.20	12.10	12.00	11.32	0.71	0.76	0.81	0.81	0.77	3.03	3.20	3.31	3.29	3.21
	DEPSE-TL	9.46	13.66	16.85	18.78	14.71	0.73	0.79	0.84	0.85	0.80	3.06	3.28	3.42	3.38	3.28
	DiffUSEEN	10.92	**14.46**	**17.23**	**18.89**	**15.39**	**0.74**	**0.86**	0.84	0.85	**0.81**	**3.09**	**3.31**	**3.42**	3.36	3.30
	ParaDiffUSE-IN	1.91	7.11	11.69	15.18	9.00	0.70	0.77	0.80	0.81	0.77	3.07	3.30	3.38	3.34	3.27
	ParaDiffUSE-EN	6.12	11.37	14.93	17.67	12.55	**0.77**	**0.82**	**0.86**	**0.86**	**0.83**	**3.19**	**3.36**	**3.47**	**3.43**	**3.36**